AI RECEPTIONIST

A 24/7 AI Phone Receptionist for Small & Mid-Sized Businesses

Version 1.0 · ai-receptionist v0.1.0 · 2026 · Confidential

24/7	< $0.05	< 1 sec	~ 6 wks
Coverage	Per Call	Response	To Launch

01 Executive Summary

A 24/7 AI phone receptionist for businesses that cannot afford to miss a call

The AI Receptionist is a self-hosted voice agent that answers every inbound call to a business, day or night, in a human-sounding voice. It greets callers, takes messages, books appointments, transfers urgent calls to staff, and answers common questions — all while the business owner sleeps, focuses on customers in front of them, or runs the business after hours. Every call is recorded, transcribed, summarised, and surfaced in a simple dashboard where staff can review activity, update what the agent says, and follow up on leads.

This Statement of Work (SOW) explains what the product does, what problem it solves, what is included, the timeline to launch, and what success looks like. It is written for non-technical readers — owners, managers, and decision-makers evaluating whether to adopt the system.

Dimension	Detail
Product	AI Receptionist — voice agent + management dashboard
Version	v0.1.0 (Scaffold complete; ready for content + tuning)
Target Customer	Service businesses: clinics, salons, legal, trades, real estate
Coverage	24/7/365 — no missed calls, no after-hours voicemail
Total Cost to Run	~$45–$90/month for ~50 calls/day
Time to Launch	~6 weeks from kickoff
Compliance	AI disclosure on every call; PCI-safe (no card data); GDPR-ready

02 Problem Statement

Why this product exists — the missed-call problem and its cost

Small and mid-sized businesses lose revenue and erode customer trust every time a phone call goes unanswered. The drivers are familiar and persistent across the service economy:

- **After-hours and lunchtime gaps.** Most SMBs cannot afford 24/7 staff; calls outside business hours go to voicemail — and 80% of callers will not leave a message.
- **Front-desk overload.** When staff are tied up with a customer in front of them, the phone rings out. The caller moves on to a competitor.
- **Slow callbacks.** Owners juggle callbacks between meetings and miss the 5-minute window when a lead is still warm.
- **Inconsistent answers.** Front-desk staff improvise on pricing, hours, and policies — leading to mistakes that cost trust.
- **No record of what was said.** Verbal commitments made on the phone vanish; disputes and follow-ups lose context.
- **Voicemail fatigue.** Callers hate voicemail and increasingly abandon the call before the beep.

The cost of a missed call

Business type	Avg. value of a missed call	Why it matters
Dental clinic	$300 – $800	New patient consults, hygiene appointments
Salon / spa	$80 – $250	Service bookings, retail add-ons
Law firm	$500 – $5,000	Case intake; one missed call can mean a lost client
Trades (plumber, HVAC, electrician)	$200 – $1,500	Emergency service calls; high urgency, high value
Real estate agent	$1,000 – $10,000+	Listing inquiries; lifetime commission value

For a typical service business, just **three missed calls per week** translates to thousands of dollars in lost revenue every month. The cost of doing nothing is rarely zero.

03 Proposed Solution

Always-on, on-brand, polite — and never tired

The AI Receptionist is an AI voice agent that sits on the business phone line. It answers immediately, identifies itself as an AI assistant (as required by law in the US and EU), and handles the call end-to-end using a configurable script tailored to the business. It uses a natural-sounding voice, understands free-flowing conversation, and recognises when a caller needs a human — at which point it transfers the call or captures the details for a callback.

Without AI Receptionist	With AI Receptionist
60% of after-hours calls go to voicemail	100% of calls answered in under a second
Front desk juggles in-person and phone — caller hears ringing	Every caller greeted warmly within ~600 ms of the line op
Owners return calls hours later, leads have moved on	Leads contacted within the configured SLA — email or SI
Staff improvise on pricing, hours, and policies	Consistent, on-brand answers drawn from a single source
No record of who called and what was said	Every call transcribed, summarised, and searchable in th
Booking only when staff are free	Bookings placed 24/7, with confirmations sent automatica

04 Scope of Work — What Is Included

Everything delivered under this SOW

The following deliverables are included in this engagement. Anything not listed here is considered out of scope and will require a separate change request.

#	Deliverable	Description
1	Voice agent	Custom-trained AI voice agent that answers inbound calls, handles common que
2	Phone number & telephony	Telnyx inbound number provisioned for the business; caller ID; routing configure
3	Calendar integration	Google Calendar integration for availability checks and 2× booking. Real-time co
4	Knowledge base	Up to 50 FAQ entries (hours, prices, services, policies, directions) — seeded du
5	Management dashboard	Web dashboard for staff to review calls, messages, and appointments; edit what
6	Notifications	Email, SMS, Telegram, and Slack notifications for new messages, bookings, and
7	Call recording & transcription	Calls transcribed automatically; AI-generated 2–3 sentence summary; searchabl
8	Daily summary report	Once-daily email summarising calls, messages, bookings, and notable escalatio
9	Self-hosting & deployment	Docker Compose deployment on the customer server (or managed hosting). TLS
10	Training & handover	Two 1-hour live sessions for staff: one on the dashboard, one on tuning the agen

05 Features

Full feature inventory, grouped by phase

Features are grouped into three release waves. Wave 1 ships with launch. Waves 2 and 3 are scheduled upgrades after launch and are included in this SOW.

Wave 1 — Launch (Phases 0–3)

Ships in ~6 weeks

- **24/7 inbound answering** — every call answered in under 1 second, with a warm, on-brand greeting
- **AI disclosure** — the agent identifies itself as an AI assistant on every call (US/EU requirement)
- **Take messages** — captures name, phone, and reason; sends to staff by email / SMS / Telegram / Slack
- **Book appointments** — checks real-time calendar availability and books the slot; sends caller an SMS confirmation
- **Transfer to staff** — for urgent calls (medical, emergency, angry customer), routes to a human phone; falls back to take-message if no answer
- **Answer FAQs** — answers from the knowledge base: hours, prices, directions, services, policies
- **Call recording & transcript** — every call recorded (if enabled) and transcribed; searchable in the dashboard
- **AI summary** — each call gets a 2–3 sentence summary plus tags (intent, outcome, sentiment)
- **Dashboard** — review calls, messages, bookings, contacts; edit what the agent says; view costs
- **Notifications** — email + SMS + Telegram + Slack; configurable per business
- **Daily report** — end-of-day summary delivered to the business owner
- **Block list** — staff can block spam numbers; blocked callers hear a brief message and are routed away
- **Spend caps** — per-call and per-day spend caps enforced automatically

Wave 2 — Intelligence Layer (30–60 days post-launch)

Updates within 60 days

- **Caller recognition** — returning callers are greeted by name and routed based on history
- **Reschedule & cancel** — callers can reschedule or cancel existing bookings by voice
- **SMS confirmations** — automatic SMS reminders 24 hours and 1 hour before appointments
- **Knowledge base growth** — staff see a list of questions the agent could not answer; one-click to add as FAQ
- **Quality review queue** — flagged calls surfaced for staff review with one-click notes
- **Cost reporting** — per-call and per-day cost dashboards; spend trends

Wave 3 — Polish & Scale (60–120 days post-launch)

Updates within 120 days

- **Outbound reminders** — the agent can place reminder calls for appointments and follow-ups
- **Spam filtering** — AI-driven detection of robocalls and spam; silent hangup

- **Multi-language** — switch between languages mid-conversation
- **CRM integration** — push contacts and call notes to HubSpot, Salesforce, or Zoho
- **Test call button** — staff can place a test call from the dashboard to validate behaviour
- **Multilingual voice** — pick voice and accent per business

06 How It Works

From caller dial to dashboard — what happens on a typical call

A typical call follows this flow. Most calls take less than two minutes.

Step	What happens	Time
1	Caller dials the business number — it rings once.	< 1 s
2	The agent answers and greets the caller by name (if recognised) and identifies itself as an AI assistant.	~ 1 s
3	The caller explains what they need — for example, "I'd like to book a haircut for tomorrow afternoon."	varies
4	The agent checks the calendar, proposes available slots, and confirms the booking.	~ 5–10 s
5	The caller receives an SMS confirmation with the appointment details.	~ 2 s
6	The call ends. Staff see a new row in the dashboard with the recording, transcript, and a one-line summary.	~ 5 s after hang-up

What the caller experiences

A warm, natural voice. No menu trees. No hold queue. No voicemail. Conversational — callers speak normally and are understood. Polite and on-brand — the agent uses the business name, the right tone, and the right words every time.

What the business owner experiences

A web dashboard where every call appears within seconds of hanging up. Each call has a one-paragraph summary, the caller's details, the recording, the transcript, and the outcome (booked / messaged / transferred). Owners tune what the agent says through a simple editor — no code required.

What staff experience

- A shared inbox of messages (with status: new → done), filtering by urgency
- A calendar view of bookings, with one-click cancel and reschedule
- A quality-review queue for flagged calls
- A contacts list with history, tags, and block-list toggle
- A settings page for users, notification channels, and retention

07 Technology Overview

A simple, modern stack — explained without jargon

The system uses a small number of proven, best-in-class vendors. The business owns its own data; nothing is locked in. All components have been selected for reliability, low latency, low cost, and ease of replacement.

Layer	Vendor	What it does for the business
Phone service	Telnyx	Carries the inbound calls to the AI agent. Low-cost, reliable, suppo
Speech-to-text	Deepgram	Transcribes the caller's voice in real time so the agent can unders
AI brain	Claude (Anthropic)	The agent's reasoning — understands the caller, decides what to
Voice (text-to-speech)	Cartesia	Speaks the agent's responses in a natural human voice. Sub-seco
Backend & API	Self-hosted (FastAPI)	Stores calls, messages, bookings, and contacts. Handles webhoo
Database	PostgreSQL + pgvector	All business data + AI-powered FAQ search.
Dashboard	Next.js (web)	The web interface staff use to review calls and manage the agent.
Notifications	Email + SMS + Telegram + Slack	Staff are notified the way they already work.
Hosting	Self-hosted Docker or managed VPS	Runs anywhere: the customer's server, a small VPS, or our mana

Why these choices

- **Self-hosted by default** — the business owns its data and its phone number. No per-minute SaaS markup; no vendor lock-in.
- **Best-of-breed** — each vendor is the strongest in its category; swapping one is a contained change.
- **Low cost** — ~$0.03–$0.06 per minute of call. A typical 2.5-minute call costs under $0.15.
- **Low latency** — typical response time is under 1 second; natural conversational feel.
- **Compliant** — AI disclosure, GDPR-ready, encryption in transit and at rest. PCI-safe (no card data ever spoken).

08 Cost to Run — What Each Call Costs

Transparent, predictable cost per call

A typical 2.5-minute call (the median for a service business) costs roughly **$0.08 to $0.15** across all providers. The breakdown:

Component	Cost per minute	Notes
Phone service (Telnyx)	~$0.008	Inbound minutes only; outbound not used in Wave 1
Speech-to-text (Deepgram)	~$0.004	Streaming partials — billed per second of audio
Voice (Cartesia)	~$0.015	Billed per character of generated speech
AI brain (Claude Haiku)	~$0.001 – $0.010	Varies with conversation length; cheaper on shorter calls
Total per minute	~$0.03 – $0.05	Most calls cluster around $0.035/min

Monthly cost examples

Call volume	Avg. length	Estimated monthly cost
~10 calls / day	2.5 min	~$25 – $40 / month
~20 calls / day	2.5 min	~$45 – $80 / month
~50 calls / day	2.5 min	~$110 – $200 / month

A simple cost-control setting caps total spend per day (default $50) — beyond which the agent politely takes a message instead of continuing. The owner is never surprised by an invoice.

09 Timeline & Delivery Plan

From kickoff to live in ~6 weeks

The project runs across five phases. Each phase ends with a live demo and a go/no-go gate before the next begins.

Phase 0	Content & Configuration	Week 1

- Discovery call — capture business hours, services, prices, policies, FAQs
- Draft the agent's persona, tone, and standard responses
- Provision phone number, calendar access, and notification channels
- Kickoff session with the business owner to confirm scope

Phase 1	Prompt Validation	Week 2

- Test the agent's responses against realistic caller scenarios (≥ 20)
- Refine the prompt until responses are consistent and on-brand
- Owner review and approval of the agent's voice and behaviour
- Capture the first baseline of agent behaviour for regression testing

Phase 2	Voice Agent Goes Live (After-Hours Only)	Week 3

- Phone number live; calls route to the AI agent outside business hours
- Live test calls covering messages, bookings, and transfers
- Email + SMS notifications working end-to-end
- Daily review of transcripts; fine-tune where needed

Phase 3	Dashboard & Prompt Editor	Week 4

- Dashboard live for staff — calls, messages, bookings, contacts
- Prompt editor live — staff can update what the agent says
- Quality-review workflow operational

Phase 4	Full-Hours Launch	Weeks 5–6

- After 1 stable week on after-hours, the agent is enabled 24/7
- Caller recognition + history enabled
- Knowledge base growth tooling live (FAQ from real questions)
- Daily summary report delivered to the owner
- Two staff training sessions (dashboard + prompt tuning)
- Written playbook handed off

10 Service Levels & Support

What we commit to after launch

Commitment	Target
Agent availability (uptime)	99.5% per month, excluding scheduled maintenance
Answer time on inbound calls	Under 1 second (95th percentile)
Notification delivery	Within 30 seconds of call end
Dashboard load time	Under 2 seconds (95th percentile)
Daily summary report	Delivered by 8:00 AM local time
Backup retention	30 days rolling (calls, transcripts, recordings)
Security patching	Critical patches within 48 hours
Bug reports — P1	Acknowledged within 4 business hours; fix or workaround within 2
Bug reports — P2	Acknowledged within 1 business day; fix scheduled within the nex
Feature requests	Reviewed monthly; roadmap update quarterly
Maintenance windows	Sunday 02:00 – 04:00 local; 7-day notice for non-emergency

11 Success Metrics

How we measure whether the system is working

Metric	Target (90 days post-launch)
Calls answered by agent	≥ 95% of inbound (excluding blocked)
Calls requiring human transfer	≤ 8% (down from baseline)
Bookings placed by agent	≥ 40% of appointment calls
Messages captured and delivered to staff	100% (within 30 seconds of call end)
Caller satisfaction (post-call survey)	≥ 4.2 / 5 stars
Average call cost	≤ $0.06 per minute
Missed-call rate	≤ 1% (down from ~60% outside hours)
Dashboard weekly active staff	≥ 2 of the business owner + 1 staff member
Staff prompt-edit frequency	Initial edits in week 1; revisions every 2 weeks

Quarterly business review

Every 90 days, we present a one-page summary to the owner: calls answered, calls transferred, bookings, spend, and the top three calls the agent could not handle — with one-click actions to update the agent's responses.

12 Out of Scope

What this SOW does not include

- **Outbound sales or marketing calls.** The agent handles inbound only in Wave 1. Outbound reminders arrive in Wave 3.
- **Handling payment card numbers by phone.** By design — the agent never accepts card numbers verbally. Payment links are sent by SMS.
- **HIPAA / protected health information.** Not supported in v0.1.0. Can be enabled with a separate compliance package.
- **Custom CRM integrations beyond the supported list.** HubSpot, Salesforce, and Zoho are supported in Wave 3. Other CRMs require a change-request.
- **Voice cloning of a specific person.** We use a curated set of natural-sounding voices. Voice cloning is a separate engagement.
- **Real-time translation between languages.** Multilingual support means the agent understands and responds in multiple languages, not simultaneous translation.
- **Hardware / phones / on-premise PBX.** We deliver a cloud-hosted or self-hosted service; we do not sell or install physical hardware.
- **Mobile apps for staff.** The dashboard is web-based and mobile-responsive in Wave 1; native mobile apps are not in scope.
- **24/7 live human support.** Email + business-hours live support in Wave 1. 24/7 live support is an add-on.

13 Assumptions & Dependencies

What we are assuming to be true at kickoff

Customer-side assumptions

- Business hours, services, prices, and policies are provided in writing within 5 business days of kickoff
- A Google Calendar is available (or willingness to create one) for booking integration
- Staff are available for two 1-hour training sessions in weeks 5 and 6
- Notification destination addresses (email, Telegram chat ID, Slack webhook) are configured before go-live
- Domain name and DNS access are available if hosted on the customer's infrastructure
- A 2 vCPU / 4 GB server is available for self-hosted deployments (or willingness to use managed hosting)

Vendor-side dependencies

- Telnyx account is provisioned with a US/CA/UK number in the business's region
- Deepgram, Cartesia, and Anthropic accounts are active with valid payment
- Google Workspace (or personal Google account) for Calendar integration
- Email delivery (SMTP or SendGrid/Resend) for outbound notifications
- SMS delivery via Telnyx messaging for confirmations

Compliance dependencies

- Recording consent: by default, calls are **not** recorded. Recording is opt-in per jurisdiction.
- Data residency: default is US-East. EU residency is available as a configuration change in Phase 4.
- GDPR data-subject requests supported within 30 days (deletion + export).

14 Acceptance Criteria

What "done" means at each gate

Gate	Done when…
Phase 0 — Content	Hours, services, prices, and FAQs documented; agent persona approved; phone num
Phase 1 — Prompt Validation	20+ realistic caller scenarios pass; tone and behaviour approved by the owner
Phase 2 — After-Hours Live	Calls answered live outside business hours for 7 consecutive days without critical bug
Phase 3 — Dashboard Live	Staff can review every call, message, and booking; staff can edit what the agent says
Phase 4 — 24/7 Launch	Agent live 24/7; staff trained; playbook delivered; first 30 days of metrics tracked and r

Final acceptance

The project is complete when Phase 4 acceptance is signed off by the business owner and the system has run cleanly for 30 days in 24/7 mode with all SLA commitments met.

15 Next Steps & Sign-off

What happens after this SOW is approved

Immediate next steps (Week 1)

	Outcome	
provision accounts	Both sides	Telnyx, Deepgram, Cartesia, Anthropic accounts ac
all (1 hour)	Vendor + Owner	Business hours, services, prices, FAQs captured
ber & calendar setup	Vendor	Inbound number live; calendar connected
persona & prompt	Vendor	Ready for Phase 1 scenario testing

Sign-off

Party	Name	Signature	Date
Customer	_____	_____	_____
Vendor	_____	_____	_____

AI Receptionist — never miss a call, never miss a customer.

This document is confidential and proprietary. Not for distribution. Companion to the Full Scope and the Technical Implementation Scope.